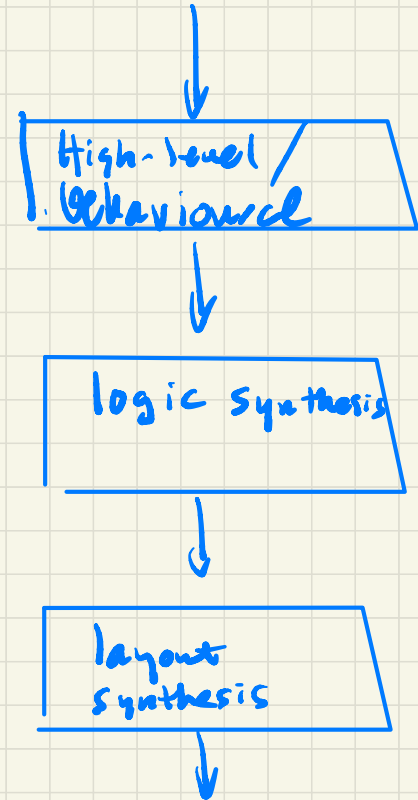


ECE 1387 Lecture #2 Sept 19, 2025

Routing Assign #1

FPGA maze routing



Routing

Generic problem statement of routing

Given: i) positions of cells (gates, FFs, or LUTs, DSP blocks)

ii) "netlist" of connections between pins on cells

Find: paths of wires so that the required connections are made;
no unrelated wires are shorted.

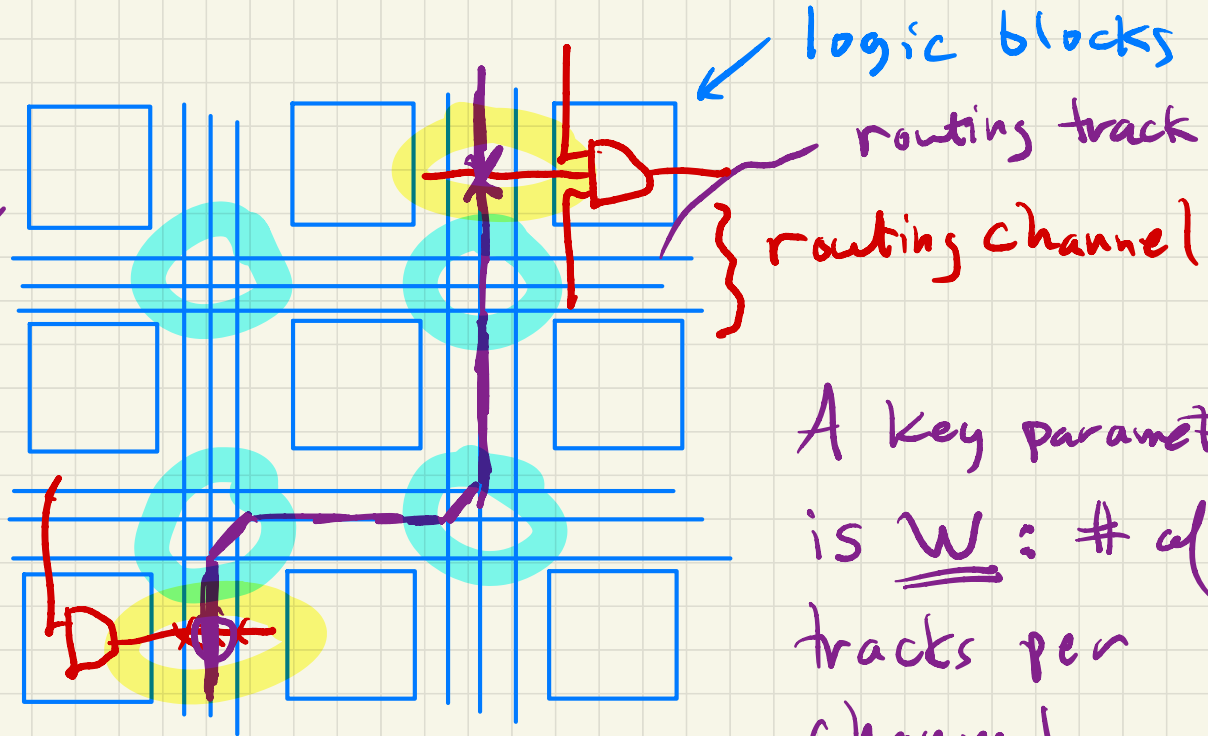
Exact problem statement depends on:

- target IC media (FPGA or ASIC)
- metal stack
- specific standard cell lib
- area target
- performance constraint
- power
- level of "detail"

Today FPGA
maze routing
→ ASIC routing
uses same
core routing
algorithm

FPGAs are an array of logic blocks surrounded by programmable routing

$W \approx 300$
in commercial
FPGA

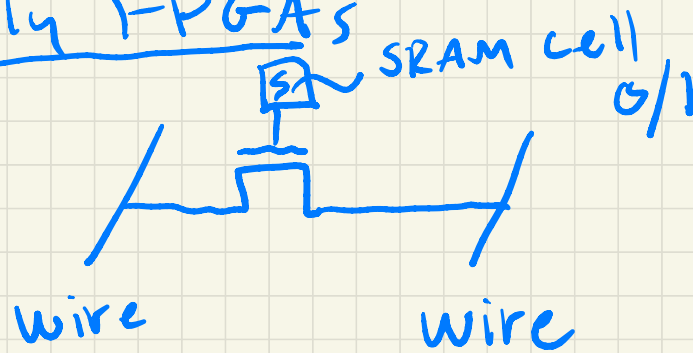


A key parameter
is W : # of
tracks per
channel -

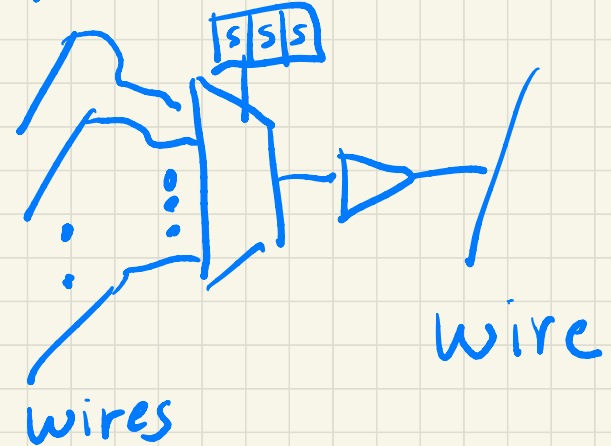
Here $W=3$

Programmable switches

Early FPGAs



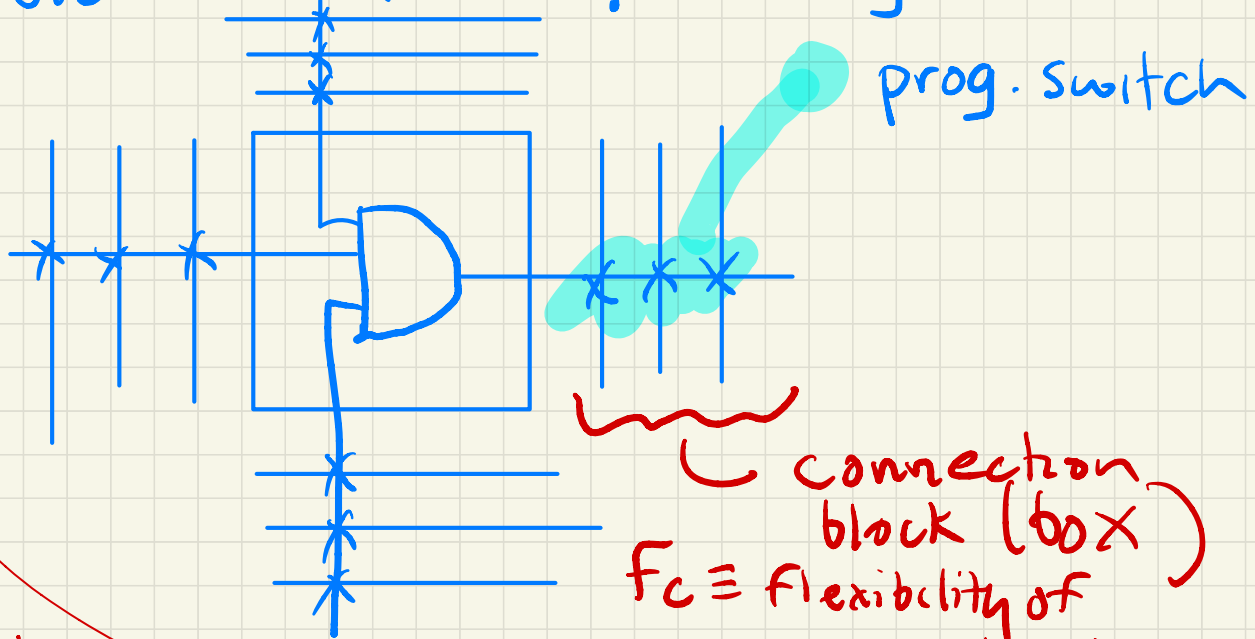
Now: FPGAs
have buffered routing
switches



A route of a connection is made by turning on the programmable switches to connect gates in desired way.

Two key routing Structures

①  Connection between pins on logic blocks and neighbouring tracks



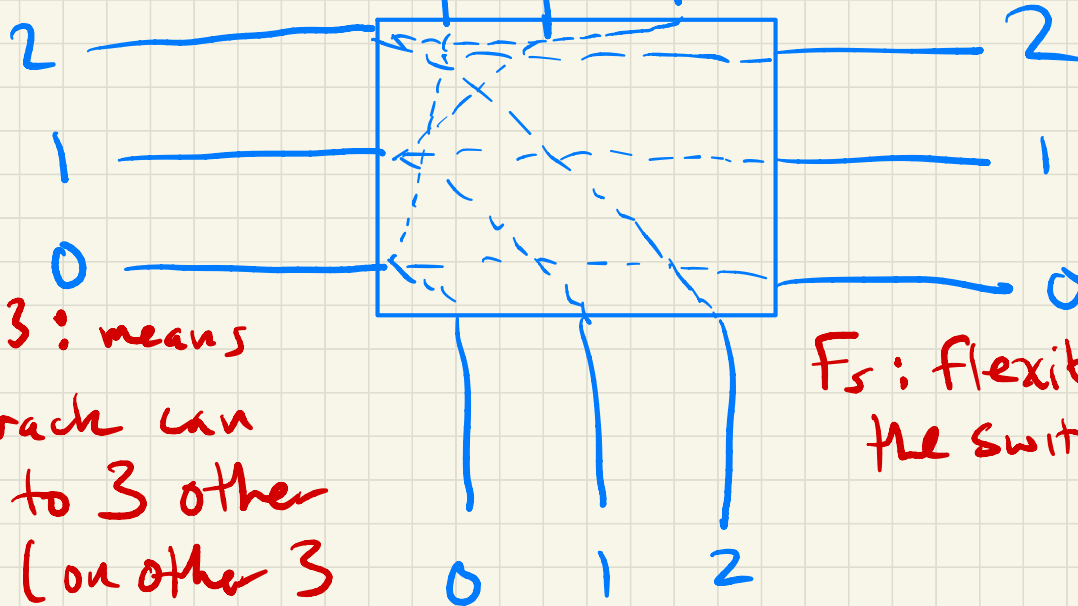
Here,
 $F_c = W$

Each pin
can connect
to all W
neighbour tracks.

② At intersection of horizontal

vertical channels

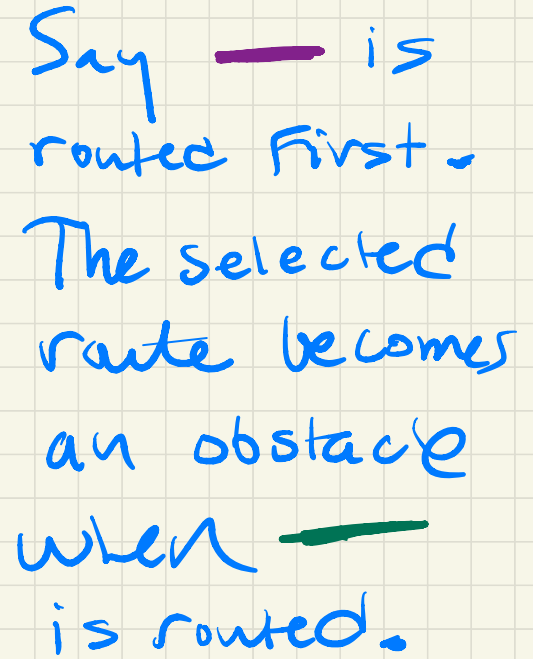
this is called
switch block (box)



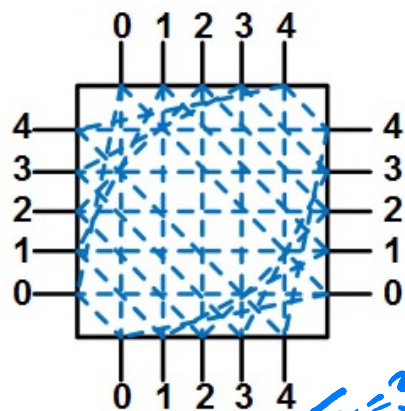
Here, $F_s = 3$: means
every track can
connect to 3 other
tracks (on other 3
sides)

F_s : flexibility of
the switch block

Turns out... switch block does not
need to be a Full crossbar



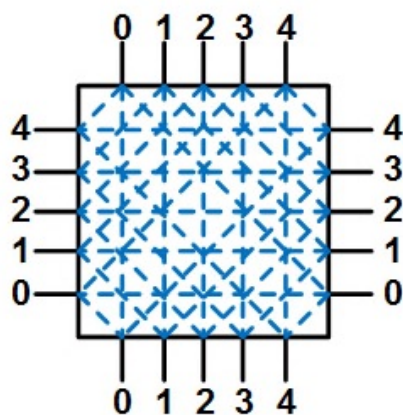
Say ~~—~~ is
routed first.
The selected
route becomes
an obstacle
when ~~—~~
is routed.



a) Disjoint

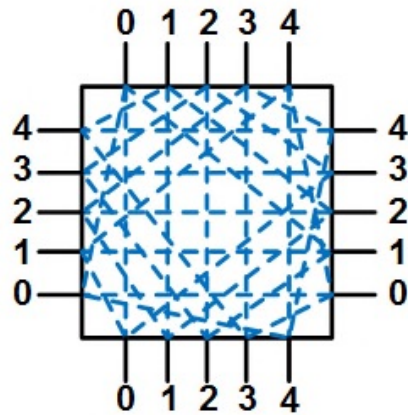
"planar"

$$F_s = 3$$



b) Universal

$$F_s = 3$$



c) Wilton

$$F_s = 3$$

Lee-Moore Maze Routing Algorithm

- Similar to Dijkstra's alg.

array? up to y.w...

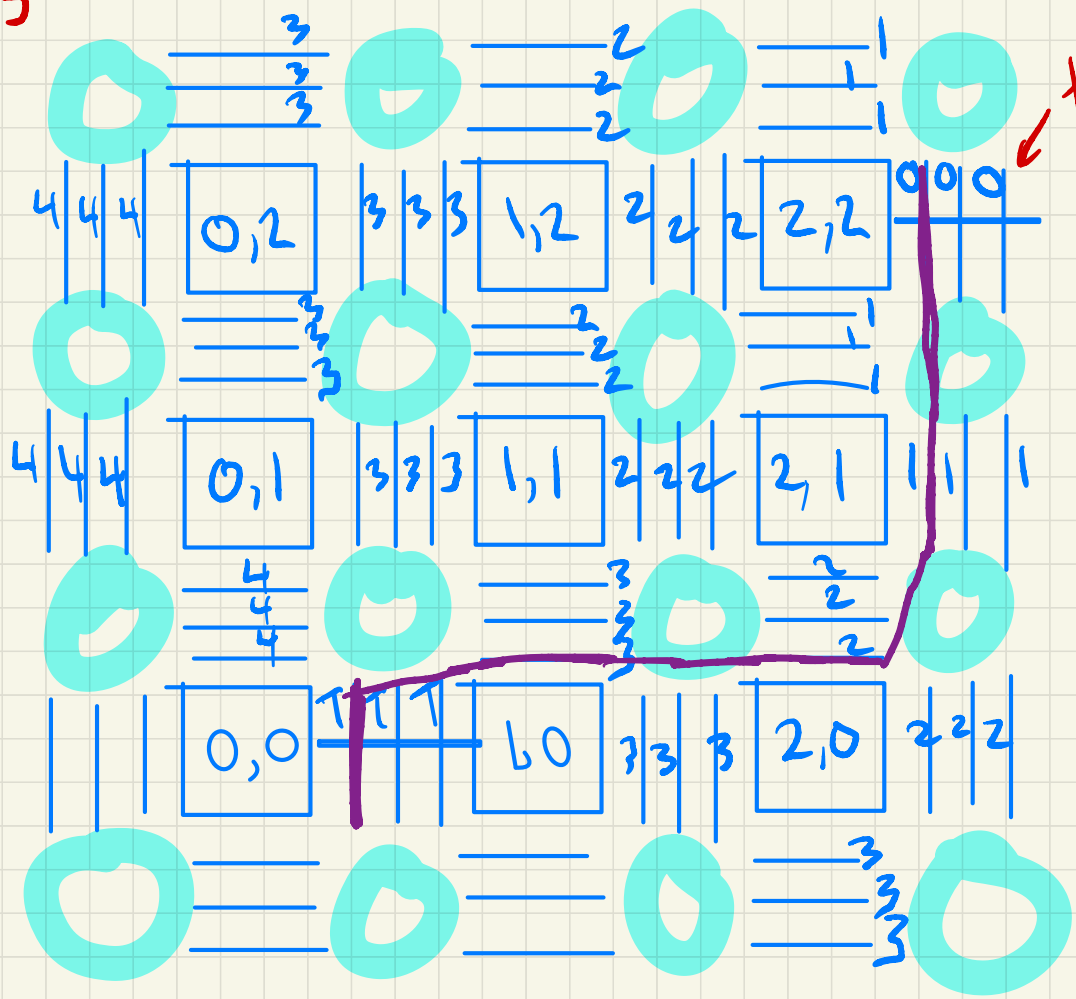
Data structure: need to have an entry (label) for each track segment.

Initially: label each track segment as avail 'A'

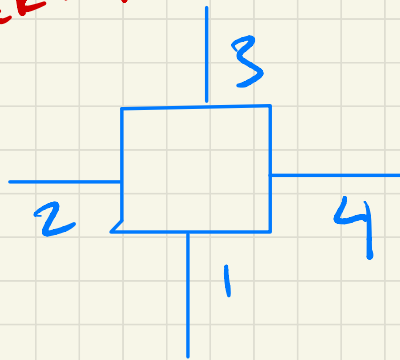
Action: route one connection at a time.

Make segment unavailable 'U'
to subsequent connections to
be routed

$w=3$



track segment



route from
2,2 pin 4 to
1,0 pin 2

For all conn's to route $\downarrow$ pin #
Source pin x_1, y_1, p_1
sink pin x_2, y_2, p_2 (target)

- 1) Mark all track segments adjacent to target 'T' (if they are avail)
- 2) Mark available track segments adjacent to source 'O' and push onto expansion list (queue)

```
while (expansion list not empty) {  
    pop segment j from expansion list  
    if j is target, exit while loop  
    for each segment k reachable from  
        j thru switch {  
        if k is available ('A') {  
            label(k) = label(j) + 1  
            push k onto expansion list  
        }  
    }  
}
```

If exit while loop having hit target
trace back thru descending
labels $\rightarrow$ define routing path for
connection $\rightarrow$ mark segments
unavailable 'U'

Reset labeled segments as 'A'

If exit while loop without hitting target $\rightarrow$ failed to route connection

This alg. will find a routing path if it exists.

This is a **heuristic** alg. Exhaustive would try all connection orderings, all trace back options,
(huge # of combinations)